

Skytree Server Release Notes

Release 15.4.0

SKYTREE®

Notices

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Customers with a support contact can contact support using any of the following methods:

- Log in to support.skytree.net.
- Send questions to support@skytree.net.
- Call 408.392.9300 and select option 3.
- You can also view additional Skytree documentation by logging on to support.skytree.net.

Freemium customers can send inquiries to the LinkedIn® Community Group for Skytree:

<https://www.linkedin.com/groups/4468542>

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Overview

Skytree is an enterprise-grade software platform for developing, testing, and deploying advanced analytic solutions for big data. With a broad set of machine learning methods, Skytree works on massive, fast changing, and varied datasets in distributed computing environments with a high degree of efficiency and accuracy.

Skytree is comprised of two components: Skytree Server™ and Skytree Platform™. These release notes describe the new features, bug fixes, and other changes made in Skytree Server.

For additional information about Skytree, refer to the documentation available on support.skytree.net.

Version 15.4.0

Version 15.4.0 provides new features, bug fixes, and other changes to Skytree Server.

New Features

- For distributed GBTR grid-search with `--regularization=off` and `--ensemble_size=0`, Skytree Server now warns if there are fewer tuning combinations than hosts, which would waste system resources, and fails with an error message if not tuning any parameter other than `--num_trees`, which cannot be distributed. In order to use multiple hosts, either add more tuning combinations, enable `--regularization`, or set `--ensemble_size` greater than or equal to the number of hosts.
- A new `--memory` option is available, which specifies the amount of memory allocated for Skytree Server computation on each host. When training, testing, or tuning via grid-search, if Skytree Server cannot operate within the given amount of memory, it fails with an error message indicating the amount of memory required for computation.
- Freemium customers can install Skytree Server on a single node. These licenses are restricted to one year and 100 million elements (Rows x Columns).

Bug Fixes

The following issues were fixed in this version of Skytree Server:

Table 1: Skytree Server 15.4.0 Bug Fixes

Issue ID	Description
CORE-1942	Fixed the way <code>link=sqrt</code> is handled during an <code>automodel</code> run when <code>glm</code> is the selected method.

Other Changes

The following updates were made in this version of Skytree Server:

Table 2: Other Changes in Skytree Server 15.4.0

Issue ID	Description
CORE-1901, 1902	Improved progress messages for status updates.
CORE-1925	Improved performance when tuning <code>cf</code> over neighbors and threshold.
N/A	Improved performance when testing and testing/traning with <code>rdfr</code> .

Version 15.3.0

Version 15.3.0 provides new features, bug fixes, and other changes to Skytree Server.

New Features

- The Generalized Linear Models Classification (GLMC) method is now available for performing machine learning on classification problems. Note that similar to SVM, GMLC models can be built from datasets that have no more than one categorical column.
- In `automodel`, `gbt`, `gbtr`, `glm`, `glmr`, and `svm (linear)`, users can now specify a `--partial_dependencies_out` option, which creates a file to store the partial dependencies of the relevant features and/or feature pairs in a JSON format. These pairs can then be used to generate partial dependence plots.
- The Skytree Server installation process now includes a built-in license file, which notifies users when their Skytree Server license expires.

Bug Fixes

The following issues were fixed in this version of Skytree Server:

Table 3: Skytree Server 15.3.0 Bug Fixes

Issue ID	Description
CORE-1819	Fixed an issue that occurred when running ensemble <code>gbt</code> in a multi-host environment and producing PMML output. The PMML file no longer writes "inf" for initial estimates in any of the trees.
CORE-1670	Fixed an issue that occurred when <code>cf</code> tuning was performed with <code>--unary=on</code> along with <code>--objective_function=hit_rate</code> . The tuning results no longer show a hit rate of 1 for all parameter combinations.

Other Changes

The following updates were made in this version of Skytree Server:

Table 4: Other Changes in Skytree Server 15.3.0

Issue ID	Description
CORE-1657	PMML export is now supported for single-host, nonlinear SVM on mixed dense-sparse data.
CORE-1723	PMML export is now available for <code>automodel</code> .
CORE-1702	PMML export files now include classification labels.
CORE-1869	Added/updated the output of Skytree Server audit files, which are available when a <code>monitor</code> path is specified in <code>skytree.conf</code> during installation. Specific changes include the following: • The directory name is now formatted <code>YYYYmmdd</code>. The individual files within the directory are now formatted <code>YYYYmmdd-HHMMSS-ffffff</code>, where <code>ffffff</code> represents the decimal microseconds of the initial timestamp. • <code>"cwd"</code> renamed to <code>"workingDirectory"</code>, which is likely the directory used when calling <code>skytree-server</code>. • <code>"abnormalExit"</code>, <code>"normalExit"</code>, and <code>"finished"</code> are combined into <code>"normalExit": false</code> until the process exits normally, with no error or signal. • Added <code>"hosts"</code>, which includes the hosts specified with the <code>--hosts</code> option or the local hostname. • Added <code>"processesPerHost"</code>, which includes the value specified with the <code>--procs_per_host</code> option or 1 if <code>--procs_per_host</code> is not given or is given as 0. • Added <code>"threads"</code>, which includes the value specified with the <code>--threads</code> option or the default if <code>--threads</code> is not given. In either case, the number is constrained both by the number of physical cores as well as <code>threads.max_threads</code> in <code>skytree.conf</code>, if present.

Version 15.2.1

Version 15.2.1 provides a new feature and bug fixes to Skytree Server.

New Features

- During installation, admins can now specify a monitor path in `skytree.conf`. This path stores audit files for individual Skytree Server runs. Refer to the *Skytree Server Installation Guide* for more information on setting a path for the `[monitor]` option.

Bug Fixes

The following issues were fixed in this version of Skytree Server:

Table 5: Skytree Server 15.2.1 Bug Fixes

Issue ID	Description
CORE-1751	Fixed an issue that caused I/O errors when using an HA NameNode for HDFS while Kerberos was disabled.
CORE-1802	When svm is the selected method during an automodel run, Skytree Server now allows the use of multiple threads during training.
CORE-1835	Improved the performance of the <code>--limit_parameters</code> option when used with <code>gbt/gbtr--smart_search</code> or when <code>--limit_parameters</code> was specified with <code>automodel</code> , and <code>gbt</code> or <code>gbtr</code> was the chosen method.

Version 15.2.0

Version 15.2.0 provides new features, bug fixes, and other changes to Skytree Server.

New Features

- The Generalized Linear Models Regression (GLMR) method is now available for performing machine learning on regression problems. Note that similar to SVM, GMLR models can be built from datasets that have no more than one categorical column.
- The `--limit_parameters` option can now be specified when running smart search on `gbt`, `gbtr`, and `svm`. Previously, this option was only available in `automodel`.

Bug Fixes

The following issues were fixed in this version of Skytree Server:

Table 6: Skytree Server 15.2.0 Bug Fixes

Issue ID	Description
CORE-1638	Fixed an issue in <code>gbt</code> that caused labels to be incorrectly computed when <code>--probabilities_out</code> was not also specified.
CORE-1628	In <code>svm</code> , <code>--pmm1_out</code> now formats sparse support vectors correctly for use with <code>skytree-evaluator</code> and other PMML readers.
CORE-1666	When running <code>svm</code> with <code>--smart_search</code> , the results now indicate whether a bias term was used.
CORE-1632	Fixed an issue that caused machine learning to fail when using a dataset that included a column with all missing values.
CORE-1626	<code>skytree-evaluator</code> now converts input data to lowercase unless the <code>-regard_case</code> option is given.

Other Changes

The following updates were made in this version of Skytree Server:

Table 7: Other Changes in Skytree Server 15.2.0

Issue ID	Description
CORE-1539	Improved efficiency of <code>gbt</code> , <code>gbtr</code> , and <code>rdfr</code> for sparse data.
CORE-1662	In non-linear SVM, attempting to specify <code>--pmm1_out</code> on combination dense-sparse data is no longer permitted.

Version 15.1.1

Version 15.1.1 provides new features and bug fixes to Skytree Server.

New Features

- In `gbt` and `gbtr`, a new `--scores_out` option can be used with `--testing_in` to output raw scores that have not been converted to labels, probabilities, or targets. Note that this cannot be used with multi-class, ranking, or ensemble `gbt/gbtr`.
- The following options were added to `gbt` and `gbtr`. Please refer to the Skytree Server User Guide for information on how to use these options:
 - `--training_offsets_in`
 - `--tuning_offsets_in`
 - `--testing_offsets_in`
- A new `--approximate_quantiles` option is available in `gbtr` when specifying a loss function of `lad` or `huber`. This flag allows an error tolerance of 0.001 for quantiles on loss functions.

Bug Fixes

The following issues were fixed in this version of Skytree Server:

Table 8: Skytree Server 15.1.1 Bug Fixes

Issue ID	Description
CORE-1542	Fixed an issue that caused multiclass <code>gbt</code> to compute incorrect results in the distributed setting.
CORE-1511	Fixed an issue that caused time values to be misinterpreted as sparse in <code>generate-header.sh</code> .

Version 15.1.0

Version 15.1.0 provides new features, bug fixes, and other changes to Skytree Server.

New Features

- PMML (Predictive Modeling Markup Language) output is now available in single-host and distributed linear svm as well as in single-host non-linear svm.
- A new `--variable_importances_out_as_json` option can be specified to write variable importances to a JSON file with feature names and column IDs (specified by `--variable_importances_out`).
- `skytree-evaluator-cli` can be used to evaluate exported PMML models against new data. This is packaged in a separate rpm: `SkytreeInfinity-evaluator-<version>.rpm`.

Bug Fixes

The following issues were fixed in this version of Skytree Server:

Table 9: Skytree Server 15.1.0 Bug Fixes

Issue ID	Description
CORE-1533	In <code>gbt</code> , users no longer receive an error when <code>--imbalace</code> is specified with <code>--regularization=on</code> .
CORE-1521	Fixed an issue that occurred when <code>--smart_search</code> was run using <code>gbt</code> with a small sampling ratio.
CORE-1506	Running a test on <code>gbt</code> no longer results in a warning message indicating that <code>--regularization</code> will be ignored.
CORE-1501	When tuning <code>gbtr</code> models using <code>--smart_search</code> , the tuning results now report the loss function type.
CORE-1531	Fixed an issue that occurred in <code>svm</code> when <code>--smart_search</code> was run with a <code>--polynomial_offset</code> value.
CORE-1457	Logistic regression now outputs integral labels with <code>--labels_out</code> .
CORE-540	Fixed an error that occurred during classification testing when the number of labels was less than the number of points.

Other Changes

The following updates were made in this version of Skytree Server:

Table 10: Other Changes in Skytree Server 15.1.0

Issue ID	Description
CORE-1499	When running <code>automodel</code> with <code>--limit_parameters=on</code> (default), Skytree Server will set the following upper bound values when <code>gbt</code> or <code>gbtr</code> is the chosen method: <code>--num_trees <= 3000</code> <code>--tree_depth <= 9</code> <code>--max_splits <= 256</code>

Table 10: Other Changes in Skytree Server 15.1.0 (continued)

Issue ID	Description
CORE-1482, CORE-1535	Improved performance and memory usage for <code>gbt</code> with sparse data.
CORE-1343	The <code>--cardinality_based_dimensions_sampling</code> option in <code>rdf</code> now defaults to <code>off</code> .
CORE-1532	<code>skytree.network.interface</code> can be specified in <code>skytree-yarn.properties</code> if a non-standard network interface name convention is used.
CORE-1157	Reduced memory usage when running <code>kmeans</code> .
CORE-1039	Improved performance of <code>gbt</code> and <code>gbtr</code> .

Version 15.0.1

Version 15.0.1 provides new features, bug fixes, and other changes to Skytree Server.

New Features

- The score module now scores recommendations written by the `--recommendations_out` file. Note that this file must be written using `--format_out=csv`. In addition, a new header line is written to the recommendations file used by `cf`. When performing recommendation scoring, the following options can be specified:
 - `--true_user_item_in`
 - `--true_ratings_in`
 - `--recommendations_in`
 - `--k_for_rec_scoring`

Note that scoring for unary recommendations now considers all true items instead of just the first k in precision/hit rate at k (in both `cf` tuning and the score module).

Bug Fixes

The following issues were fixed in this version of Skytree Server:

Table 11: Skytree Server 15.0.1 Bug Fixes

Issue ID	Description
CORE-1321	When tuning and training with <code>gbt</code> , <code>-nan</code> and <code>-inf</code> no longer display as values in the tuning results output.
CORE-1477	In <code>gbt</code> , fixed an issue that resulted in models being trained during tuning even when <code>--model_out</code> and <code>--testing_in</code> were not specified.
CORE-1199	In <code>gbtr</code> , users now receive an error message if they attempt to specify <code>--ensemble_size</code> along with <code>--regularization=on</code> .
CORE-1495	Specifying <code>--pmm1_out</code> during <code>gbt</code> training no longer fails on MapR.

Table 11: Skytree Server 15.0.1 Bug Fixes (continued)

Issue ID	Description
N/A	Fixed an error that occurred when using <code>--smart_search_restart_out</code> with multiple processes on Hadoop.
CORE-1166, CORE-1433	When running <code>automodel</code> with the <code>--model_in</code> option, the log file no longer returns a message indicating that the model can be reproduced using a specific <code>--smart_search_seed</code> . A smart search seed will only be returned if <code>automodel</code> is run with <code>--restart_in</code> and without <code>--model_in</code> .
CORE-1446	In <code>skytree-yarn</code> on MapR, fixed an error that occurred when specifying an absolute path for <code>--output</code> .
CORE-1447	In <code>skytree-yarn</code> , the <code>--filter-stdout</code> option no longer filters out the command line from the yarn app output.

Other Changes

The following updates were made in this version of Skytree Server:

Table 12: Other Changes in Skytree Server 15.0.1

Issue ID	Description
CORE-1460	<code>--pmm1_out</code> is now supported for multi-class classification in RDF and GBT.

Version 15.0.0

Version 15.0.0 provides new features, bug fixes, and other changes to Skytree Server.

New Features

- PMML (Predictive Modeling Markup Language) output is available for `gbt`, `gbtr`, `rdf`, and `rdfr` models using the `--pmm1_out` option. The output allows you to use external code for scoring. Note that the PMML file produced requires that missing values are encoded as '?', which is a reserved symbol.
- Smart search support has been extended to `gbtr` and `svm`.
- Smart search now reports multiple scoring metrics rather than scoring only based on the testing objective.
- The `predict` module has been renamed `automodel`. In addition, the `automodel` module can now tune, train, and test regression models.
- A new `--limit_parameters` option is available in `automodel`. When enabled, this flag restricts the parameter space (for example, in `--num_trees`, `--loss_function`, `--lambda`, etc.) in order to reduce the training time.
- Reduced peak memory on the master node in a multi-host (distributed) execution of `gbt` and `gbtr`.
- The `gbtr` and `rdfr` modules now accept a `--training_point_weights_in` option for updating points weights when scoring regression models.

- skytree-yarn now accepts a `--filter-stdout` option that improves readability of log messages.
- The `--visualization_out` option can now be used with models that were trained with datasets that included missing values.
- skytree-yarn is now supported on MapR 4.0.2. To use skytree-yarn on MapR 4, update the `skytree-yarn.properties` for MapR as follows:
 - Add the path to `mapr.login.conf`. This is usually in `/opt/mapr/conf`.
`skytree.jass.config.path=<path/to/mapr.login.conf>`
 - Add the MapR NFS mount path. This is usually in `/mapr/<cluster_name>`.
`skytree.mapr.nfs.mount.path=</maprfs/nfs/mount/path>`
- skytree-yarn now writes a JSON log of each execution when `skytree.monitor.dir` is specified in `skytree-yarn.properties`.
- A new `--yield_values_in` option allows you to specify a file to be used to calculate yield values during tuning. In addition, `yield` is now an available parameter that can be used with `--testing_objective`. When specified along with the `--yield_values_in` option, the yield value will be used to determine the best model.
- Two new options are available in the classification score module: `--pr_curve_out` and `--pr_curve_size`. These options allow a confusion matrix to be available with an accompanying precision recall (PR) curve.
- Singular values and vectors now output to HDFS in `.st` format. This allows `svd` to be used in dimension reduction before a supervised machine learning algorithm and, thus, used in predictive maintenance.
- Skytree Server now allows libsvm-formatted sparse data. A new `-num_sparse_input_columns` option in `generate-header.sh`, `etl-hadoop.sh`, and `time-slicer.sh` allows you to specify the number of sparse columns that are included in the libsvm-formatted data. In addition, `convert-data.sh` accepts sparse data for headers that were prepared using this option.

Bug Fixes

The following issues were fixed in this version of Skytree Server:

Table 13: Skytree Server 15.0.0 Bug Fixes

Issue ID	Description
CORE-1218	Fixed an error in <code>gbt</code> that occurred when <code>--loss_function=ndcg</code> was specified.
CORE-1316	Fixed an issue that occurred when tuning <code>gbt</code> in non-ensemble, distributed mode. In some cases, tuning in parallel resulted in duplicate data in the tuning table.
CORE-1182	In <code>gbtr</code> , fixed a scoring issue that occurred while running on multiple hosts with an explicit tuning table and with regularization turned on.
CORE-1119, CORE-1168	Fixed a failure that occurred while running <code>gbt</code> smart search or <code>automodel</code> when testing with multiple hosts.
CORE-1134	In <code>gbt</code> and <code>gbtr</code> , reported model parameters now show whether regularization is turned off when training a model. Prior versions did not record this value, which made it difficult to reproduce a model.

Table 13: Skytree Server 15.0.0 Bug Fixes (continued)

Issue ID	Description
CORE-1150	The <code>regularization</code> parameter in <code>gbt</code> and <code>gbtr</code> can now be set implicitly (i.e, without requiring " <code>=on</code> ").
CORE-1392	In <code>gbtr</code>, an error is now presented to the user if the following loss function configurations are attempted: - Specifying <code>--huber_loss_quantile</code> without specifying <code>--loss_function=huber</code> - Specifying <code>--tweedie_exponent</code> without specifying <code>--loss_function=tdlog</code> - Specifying <code>--log_clamp</code> without specifying <code>--loss_function=pslog</code>, <code>--loss_function=gmlog</code>, or <code>--loss_function=tdlog</code>
CORE-1165	Fixed an issue that used the absolute value of Gini when tuning.
CORE-1393	In <code>kmeans</code> , an error is now presented to the user if <code>--output_with_ids</code> is specified without also specifying <code>--memberships_out</code> and/or <code>--distortions_out</code> . Users will also see an error if <code>--memberships_out</code> or <code>--distortions_out</code> is used along with <code>--kmin</code> and <code>--kmax</code> .

Other Changes

The following updates were made in this version of Skytree Server:

Table 14: Other Changes in Skytree Server 15.0.0

Issue ID	Description
CORE-1050, CORE-1114	Improved tree construction in stochastic <code>gbt</code> and <code>gbtr</code> .
CORE-1071	Improved speed when performing <code>gbtr</code> with <code>--regularization=off</code> while training models on a single machine.
CORE-1163	Improved performance of <code>rdfr</code> .
CORE-1185	In <code>gbt</code> , <code>predict</code> , <code>rdf</code> , and <code>svm</code> , the <code>--testing_objective=yield_dev</code> option has been replaced with <code>--testing_objective=yield</code> .
CORE-562	The <code>--reward_col_idx</code> and <code>--reward_table_in</code> options have been removed in favor of the <code>--yield_values_in</code> option.
CORE-1358	Improved the probability calculation when performing stratified sampling in <code>gbt</code> .
CORE-199	Improved performance of single-machine and distributed linear <code>svm</code> .
CORE-1377	In <code>skytree-yarn</code> , the <code>--mem</code> option is deprecated and is replaced with the <code>--memory</code> option. Note that <code>--mem</code> will be removed in a future release.
CORE-1153	A <code>-delimiter</code> option is now available in <code>time-slicer.sh</code> .

Table 14: Other Changes in Skytree Server 15.0.0 (continued)

Issue ID	Description
CORE-1153	The <code>-delimiter</code> option now allows a SPACE (or <code>'</code>) delimiter. Note: Because multiple spaces can be used to separate columns, it is necessary to explicitly denote empty column values with <code>""</code> when using <code>-delimiter=SPACE</code>.

